02-19-2025

HW due Wednesday:

VFI:

$$v^{(n+1)}(k_i) = \max_{\substack{k_j \\ \in \\ \{k_1,\dots,k_N\}}} \{u(zf(k_i)+(1-\delta)k_i - k_j) + \beta v^{(n)}(k_j)\}$$

↳ Howard Improvement Algorithm:

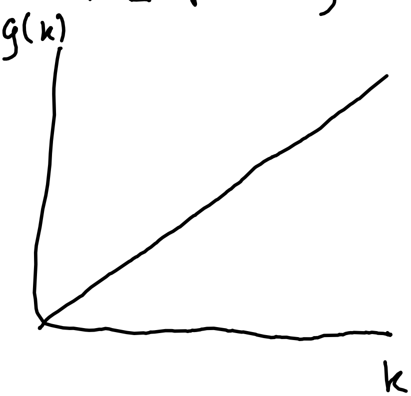
→ basically, skip "max"

$$v^{(n+1)}(k_i) = u(zf(k_i)+(1-\delta)k_i - g^{(n)}(k_i)) + \beta v^{(n)}(g^{(n)}(k_i))$$

plug in previous period's policy function

↳ must be used with value iteration as not to deviate

↳ Exploring Monotonicity of Policy Function



↳ at iter. i, $j_i^* = \underset{j\in\{1,\dots,n\}}{\arg\max}\ \Pi_{i,j}$

↳ at i+1: $j_{i+1}^* = \underset{j\in[j^*_{i-2},\dots,n]}{\arg\max}\ \Pi_{i+1,j}$

step back, but don't go over whole distribution i.e. if at i=80, start with i+1=78

VFI w/ uncertainty:

$$V(k,z) = \max_{k'} \{ u(e^{z} f(k) + (1-\delta)k - k') + \beta \sum_{\gamma=1}^{m} V(k, z)$$

$$z = \{ z_1, \dots, z_m \} \; ; \; k = \{ k_1, \dots, k_n \}$$

$V^{(0)}(1:n, 1:m)$

$k \; , \; z$

$$T^{(\gamma)} = \begin{bmatrix} T_{11}^{(\gamma)} & \cdots & T_{1n}^{(\gamma)} \\ \vdots & \ddots & \vdots \\ T_{m1}^{(\gamma)} & \cdots & T_{nn}^{(\gamma)} \end{bmatrix}$$

$$T_{ij}^{(\gamma)} = u(e^{z_\gamma} f(k_i) + (1-\delta)k_i - k_j) + \beta \sum_{s=1}^{m} P_s V^{(0)}(j, s)$$

$k' = g(k, z) \Rightarrow$ depends on each new z

$\hookrightarrow z$ is fixed

Example:

$$v(a,z) = \max_{a'} \{u(a(1+r)+wz-a') + \beta E v(a',z')\}$$

$a \in [1,n]$; $z \in [1,n]$, $P_{h,r} = Pr(z' = z_h | z = z_r)$

$V_1 = \{v(a_1, z_1), v(a_2, z_1), \dots, v(a_n, z_1)\}$

$V_2 = \{v(a_1, z_2), v(a_2, z_2), \dots, v(a_n, z_2)\}$

$$\underset{n\times 1}{\mathbb{1}} = \begin{pmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{pmatrix}$$

use if to handle negative
assign negative to value to make positive in MatLab

$$\underset{n\times n}{R_1}(i,j) = u(a_i(1+r) + wz_1 - a_j)$$

$$\underset{n\times n}{R_2}(i,j) = u(a_i(1+r) + wz_2 - a_j)$$

$$T(v) = \max(\dots + \beta v)$$

$$\to tv_1 = \max\{\underset{n\times n}{R_1} + \beta P_{11} \underset{n\times 1}{\mathbb{1}} \underset{1\times n}{v^{(0)'}(:,1)} + \beta P_{12} \underset{n\times 1}{\mathbb{1}} \underset{1\times n}{v^{(0)'}(:,2)}\}$$

After applying max, get vector back

$\hookrightarrow$ either $1\times n$ or $n\times 1$

$$\underset{n\times 1}{\overset{v^{(1)}}{\overset{\shortparallel}{tv_1}}} = \max\{\underset{n\times n}{R_1} + \beta P_{11} \underset{n\times 1}{\mathbb{1}} \underset{1\times n}{v^{(0)'}(:,1)} + \beta P_{12} \underset{n\times 1}{\mathbb{1}} \underset{1\times n}{v^{(0)'}(:,2)}\}$$

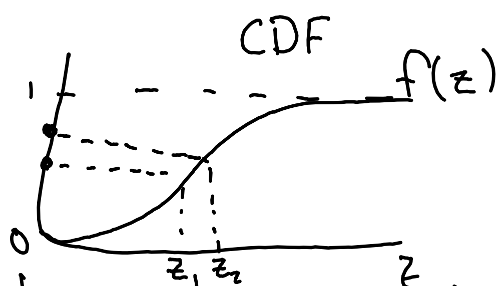
$$(i,j) \Rightarrow \underbrace{u(a_i(1+r)+wz_1 - a_j)}_{R_1(i,j)} + \beta P_{11}(j,1) + \beta P_{12}(j,2)$$

$$v^{(1)}(:,2) = tv_2 = \max\{R_2 + \beta P_{21} \mathbb{1} v^{(0)'}(:,1) + \beta P_{22} \mathbb{1} v^{(0)'}$$

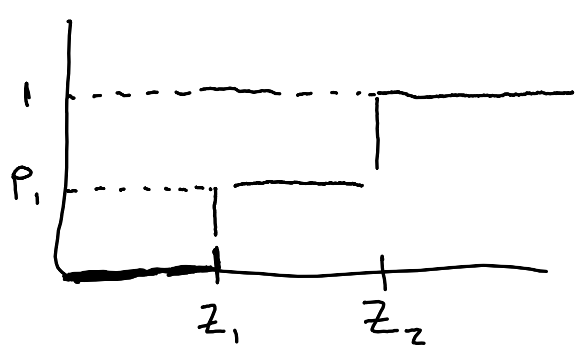
then update in loop and repeat

Simulations:

$Z \in [\underset{p_1}{z_1}, \underset{1-p_1}{z_2}]$



command rnd draws random number b/w 0-1



if RND $\leq$ P1;
assign $z = z_1$
if RND $>$ P1;
assign $z = z_2$

ctime $\doteq (1, \ldots, 1000)$ consumption

atime $(1, \ldots, 1000)$ asset

$t=1$; $z_1 = 0$ init assets $= a_0$

$a' = p(a_0, z_1)$

atime$(1) = a'$

ctime$(1) = a_0 + wz_1 - a'$

$t=2$; z_2

$a' = g(\text{atime}(1), z_2)$

atime$(2) = a'$

ctime$(2) = \text{atime}(1)(1+r) + wz_2 - a'$

$\vdots$

drop first several obs. in infinite horizon

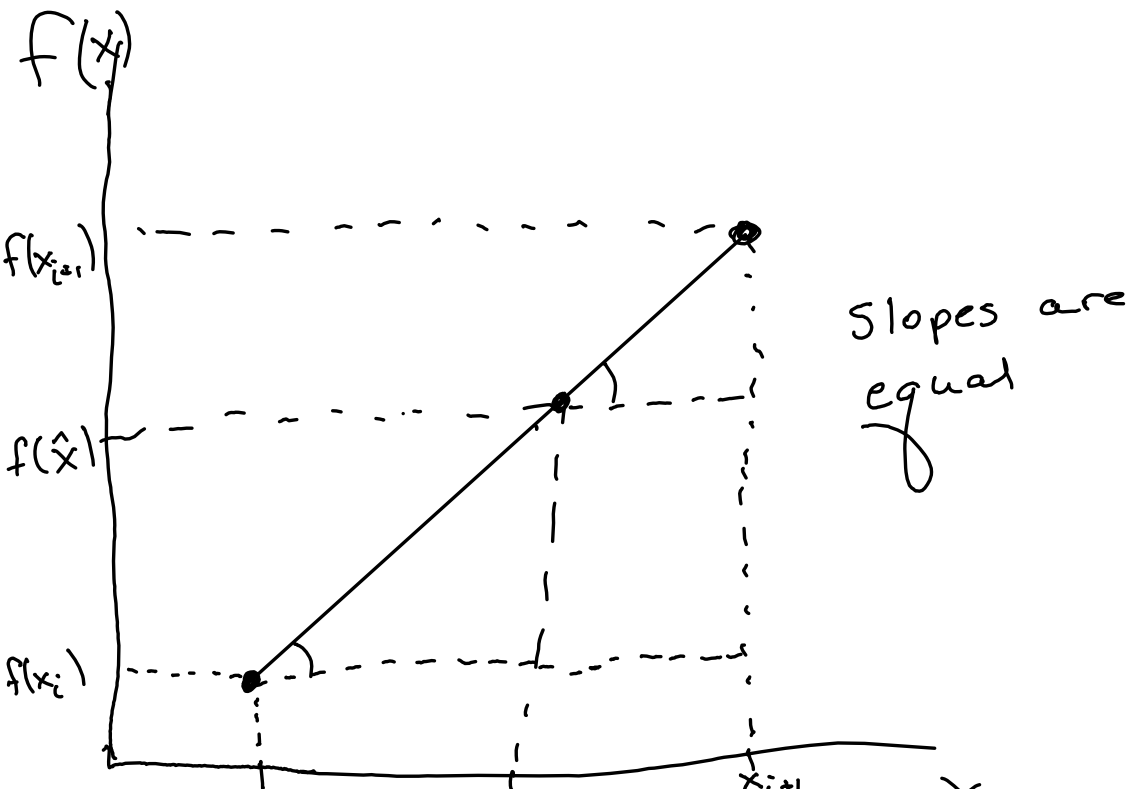
$u(zf(k_i) + (1-\delta)k_i - k') + \beta v^{(0)}(k')$

Linear Interpolation:

$x_1 \quad x_2 \quad \ldots \quad x_n$

$f(x_1) \quad f(x_2) \ldots f(x_n)$

$\hat{x} \Rightarrow f(\hat{x})$ outside of above set



find points $\hat{x}$ is between $x_i < \hat{x} < x_{i+1}$

↳ find in MatLab